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PUBERTAL DISORDERS

Puberty is defined by the appearance of secondary sexual characters.

First change in puberty $\left\{ \begin{array}{l} \rightarrow \text{In females : Thelarche (Occurs earlier than males).} \\ \rightarrow \text{In males : Increase in size of testis.} \end{array} \right.$

Delayed puberty

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Definition :

In male : Pubertal changes does not occur > 14 yrs of age.

In female : $\left\{ \begin{array}{l} \rightarrow \text{Absence of thelarche } > 13 \text{ yrs of age} \\ \rightarrow \text{Absence of menarche by } > 16 \text{ yrs of age OR } > 5 \text{ yrs after thelarche.} \end{array} \right.$

ETIOLOGY :

Overall m/c cause : **Constitutional delay** in growth and puberty
(Also m/c cause of short stature).

Central/hypogonadotropic hypogonadism :

- D/t pituitary gland and/or hypothalamus disorder.
- $\downarrow$ LH, FSH $\rightarrow$ $\downarrow$ sex steroids.

1. Syndromes :

- **Kallmann** : Associated with anosmia.
- **Prader-willi** (Genomic imprinting disorder) :
- Obesity, almond eyes, small hands.
- **Laurence-moon-Biedl-Bardet** :
- Obesity, polydactyly, retinitis pigmentosa.

2. CNS injury :

- Trauma.
- Infiltrative disorders of pituitary :
Histiocytosis, sarcoidosis, etc.

khanirfan0392@gmail.com

Peripheral/hypergonadotropic hypogonadism :

D/t $\downarrow$ sex steroids $\xrightarrow{\text{Feedback}}$ $\uparrow$ LH, FSH.

1. Chromosomal anomalies :

- Turner syndrome : Streak ovaries.
- Klinefelter syndrome :
Cryptorchidism $\rightarrow$ $\downarrow$ testosterone production.

2. Congenital adrenal hyperplasia (CAH) :

$\downarrow$ 17- α -hydroxylase $\rightarrow$ Delayed puberty in males

3. Insufficient androgen action :

- Androgen insensitivity syndrome (AIS).
- 5- α -reductase deficiency.

ANDROGEN INSENSITIVITY SYNDROME (AIS) :

Pathology : Resistance to androgen action.

Genetic defect : AR gene mutation.

mode of inheritance : X-linked recessive.

Karyotype : 46 + XY.

Phenotypic appearance of classical form (Features vary with severity of resistance).

- Female.
- Breast development : AIS $\rightarrow$ $\uparrow$ levels of testosterone aromatase $\rightarrow$ estrogen
Acts on $\rightarrow$ breast tissue.
- uterus : Absent.
- Pubic and axillary hair : Absent/spare.
- Undescended testis : Palpable inguinal mass.

5- α -REDUCTASE DEFICIENCY :

Pathophysiology :

Testosterone $\xrightarrow{\text{5-}\alpha\text{-reductase}}$ 5-dihydrotestosterone (5-DHT) $\rightarrow$
 $\downarrow$ development of male external genitalia (Fusion of scrotum, lengthening of penis).

Clinical features :

- Female-like features : Child raised as a girl.

$\downarrow$
 Puberty : $\uparrow\uparrow$ testosterone levels
 $\downarrow$

- Spontaneous virilization.
- Apparent change of gender.

MANAGEMENT :

Replacement of sex steroids :

In females :
 $\downarrow > 12y$
 Ethinylestradiol
 +

In males:
 $\downarrow > 14y$
 Testosterone

Oral progesterone 2 yrs later
 (To achieve break-through bleeding)

- Early administration $\rightarrow$ Early fusion of epiphysis $\rightarrow$ Short stature.

Precocious Puberty

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Pubertal changes occurring in — { Females >5y
males <9.5y

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> Peripheral/gonadotropin independent:

1. Functioning tumors :

more common in testis $\rightarrow \uparrow$ testosterone production.

- Leydig/sertoli cell tumors.

2. Autonomously functioning ovarian cyst: **McCune**

Albright syndrome.

- ↑ production of estrogen.

3. Congenital adrenal hyperplasia:

Deficiency of $\left\{ \begin{array}{l} \text{21-hydroxylase} \\ \text{11-}\beta\text{-hydroxylase} \end{array} \right\}$ Precocious puberty in males.

Autonomous endocrine hyperfunction of one or more of the following :

- Pituitary.

- Adrenal gland.

- Ovaries.

Genetic defect: **GNAS1 gene mutation** (Gain of function) → Autonomous uncontrolled receptor activation.

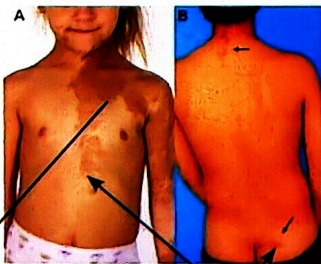
↓ Codes for
α-subunit of Gs receptors

polyendocrinopathy
(precocious puberty)

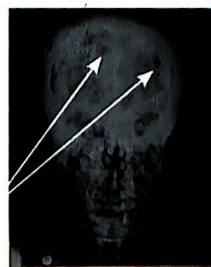
Cafe-au-lait spots
(Skin finding) Δ

Triad:

Polyostotic fibrous dysplasia: multiple lytic lesions in bones.



Lytic lesions
in skull



m/c sites affected:

1. Base of skull.

2. Proximal femur $\rightarrow$ Bowing

Shepherd's
crook
deformity

Irregular margins :
"Coast of maine"
appearance

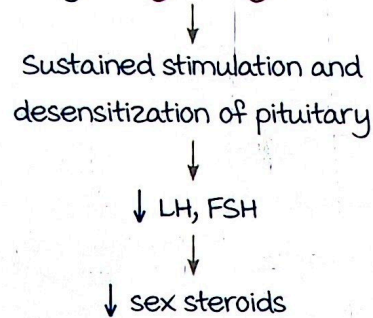
Spots do not cross
midline

Note : Café-au-lait spots in neurofibromatosis-1 (NF-1) have regular margins "Coast of California" appearance.

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TREATMENT :

- Treatment of underlying condition.
- medical management :
 - McCune Albright syndrome : Aromatase inhibitors (Eg : Letrozole) $\rightarrow$ $\times$
 - Final step of estrogen synthesis $\rightarrow$ $\downarrow$ estrogen.
 - Central precocious puberty : Long-acting GnRH analogue (Eg: Leuprolide).



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